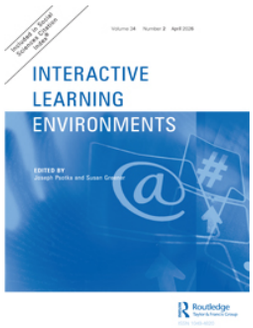

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How teachers' digital literacy influences the intention to use AI teaching tools: an empirical study based on an integrated model

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ABSTRACT
The rapid development of artificial intelligence (AI) has led to widespread use of intelligent teaching assistants and personalized learning platforms. Teachers' digital literacy (DL) plays a critical role in their intention to use (ITU) such tools. This study integrates the Technology Acceptance Model (TAM) and Theory of Planned Behavior (TPB) to explore how DL influences ITU through perceived usefulness (PU), behavioral attitude (BA), and subjective norms (SN). Using partial least squares structural equation modeling (PLS-SEM) and multi-group analysis (MGA), we examined gender and teaching experience differences. Results show that DL significantly impacts ITU both directly and indirectly via PU, BA, and SN. PU and BA serve as a chain mediating mechanism. The model accounts for 59.5% of the variance in ITU. Notably, the analysis revealed gender differences in the path from PU to ITU. These findings offer theoretical and practical implications for promoting AI adoption in education, especially in teacher training and support.

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AI; Digital literacy; TAM; TPB; intention to use; multi-group analysis (MGA)

1. Introduction

As technology progresses rapidly, AI is being more widely implemented in numerous sectors, particularly in education, where it is transforming traditional teaching models and learning approaches. As pivotal participants in the educational process, teachers' acceptance of AI teaching tools directly impacts the practical effectiveness of these tools and the improvement of educational quality. As a result, gaining insights into and encouraging educators' adoption and use of AI-based teaching tools has become a crucial focus in current educational research.

AI is an evolving field that combines knowledge from various disciplines and utilizes computational tools to explore and create frameworks, strategies, technologies, and applications designed to replicate, enhance, and broaden the scope of human cognitive abilities (Liu et al., 2021). At present, AI is widely used in education, significantly improving efficiency and supporting tailored learning experiences (Puiu & Udrițioiu, 2024). AI tools like smart tutoring systems, intelligent assessments, and virtual teaching assistants enhance teaching efficiency and student engagement (Borger et al., 2023; Xiao & Zhi, 2023). However, despite the numerous advantages of AI teaching tools, their practical application still faces several challenges. Teachers are pivotal in shaping how these tools are embraced and applied within the learning environment, which is vital for their successful implementation. Therefore, investigating teachers' acceptance and intention to use (ITU) AI teaching tools holds significant practical importance.

Amid the advancements of the modern digital world, mastering digital literacy (DL) has become a fundamental ability, critical for both routine activities and career advancement. Recent studies mainly explored how DL influences technology adoption intentions, especially in educational technology and AI applications (Kabakus et al., 2023; Marques & Sabota, 2020). Kabakus et al. (2023) explored how DL of higher education administrators impacts their ITU digital tools in task execution, revealing a strong relationship between their DL skills and their ITU digital tools. More recently, some studies have adopted TAM and TPB to examine how DL influences individuals' attitudes toward adopting new technologies (Feng & Sumettikoon, 2024; Liang et al., 2024; Lu, Sun, et al., 2024b; Ullah et al., 2022; Yao & Wang, 2024). Yao and Wang (2024) evaluated factors influencing the willingness of future special education teachers to

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incorporate AI into their teaching practices, showing that DL positively impacted perceived usefulness (PU) and perceived ease of use (PEOU), which in turn influenced their ITU AI-assisted teaching tools. While extensive research has examined the mechanisms through which DL affects ITU, studies leveraging TPB to explore this relationship are still limited. Furthermore, research specifically focusing on teachers as the target population remains relatively scarce. In addition, prior studies examining how DL influences the ITU of AI-based teaching tools tend to neglect the distinct attributes of different groups, such as gender and professional teaching background. In light of the potential differences between gender and teaching experience groups, this study further employs MGA using SmartPLS to evaluate variations in teachers' ITU AI teaching tools.

Therefore, this study focuses on addressing two key theoretical gaps in the application of AI within educational contexts. First, while prior studies have examined how DL relates to the adoption of technology, theoretical investigations into how DL, as a critical antecedent variable, influences teachers' ITU AI teaching tools in educational settings remain scarce. This has constrained our comprehension of the fundamental processes that influence teachers' ITU of new technologies. Second, prior studies have rarely examined the moderating effects of teachers' individual background differences within technology adoption models, particularly regarding how gender and teaching experience influence the pathways through which various variables affect ITU in integrated models of TAM and TPB. This has constrained the applicability of existing theoretical models in explaining heterogeneous adoption behaviors among teachers. In response, this study combines TAM and TPB, incorporates digital literacy as a key antecedent factor, and thoroughly investigates how gender and teaching experience moderate the relationships, aiming to offer a deeper and more holistic insight into the factors shaping teachers' acceptance of AI teaching tools. This study contributes to expanding the theoretical foundations related to the adoption of educational technologies and provides novel conceptual viewpoints along with actionable recommendations for facilitating the successful incorporation of AI into teaching practices. Therefore, the present research seeks to address these identified gaps by exploring the following questions:

1. Does DL directly or indirectly influence teachers' ITU AI teaching tools?
2. How well does the conceptual model put forward account for the variation in teachers' ITU AI teaching tools?
3. Which constructs mediate the influence of DL on ITU in this study?
4. Do gender and teaching experience moderate the influence of DL on teachers' ITU AI teaching tools?

2. Literature review and research hypotheses

2.1. Literature review

2.1.1. TAM

According to Davis (1989) introduced TAM, the decision to embrace new technologies is shaped by PU and PEOU, which positively predict their intention and behavior to adopt such technologies. In TAM, PU indicates how much a user is convinced that employing a specific technology can boost their work efficiency, while PEOU reflects the level of effort necessary to interact with the technology. The ITU represents the probability or planned effort of an individual to undertake a specific action in the future (Davis, 1989).

Early validations of TAM primarily focused on the information technology sector, particularly in the adoption of computer software. Over time, the model's application has expanded to fields such as educational technology (Antonietti et al., 2022; Sprenger & Schwaninger, 2021). As an example, Antonietti et al. (2022) applied TAM to explore how teachers' perceptions of DL are linked to their intention to incorporate educational technologies, explaining 73.4% of the variance in ITU.

Some prior works have built upon TAM by adding contextual dimensions – such as trust perception, risk awareness, enjoyment in use, DL, and individuals' potential for innovation – to enhance its explanatory power (García-Alonso et al., 2024; Pan, Xie, et al., 2024; Sahin et al., 2024; Wu & Zhang, 2024; Xu et al., 2023). These efforts aim to offer a more thorough investigation of how these factors influence users' cognitive attitudes, ITU, and actual usage behavior. For instance, Pan, Xie, et al. (2024) proposed an adapted TAM model to analyze the critical determinants shaping university students' willingness to engage with AI-driven

coding support systems. Their findings indicated that perceived trust and perceived risk significantly impacted PU, PEOU, and ITU.

Researchers have also integrated TAM with other theoretical models, such as TPB, Expectation Confirmation Model, Task-Technology Fit Model, and Unified Theory of Acceptance and Use of Technology (AlHadid et al., 2022; Prikshat et al., 2024; Rafique et al., 2024; Tawafak et al., 2023; Wang et al., 2022). For example, Prikshat et al. (2024) applied a unified model incorporating TPB and TAM to explore human resource professionals' ITU AI-enhanced human resources management systems. Their findings highlighted that PEOU positively influenced PU, which significantly influenced ITU.

In summary, TAM has been widely applied in predicting users' ITU new technologies. Accordingly, this study leverages TAM to analyze the elements affecting educators' ITU AI teaching tools.

2.1.2. TPB

TPB, proposed by Ajzen (1991), suggests that an individual's behavior may be anticipated based on their BA and the associated SN (Sheeran et al., 2003). According to TPB, BA refer to a person's positive or negative evaluation of performing a specific behavior, while SN reflect the perceived social pressure from significant others regarding whether the behavior should be carried out (Ajzen, 1991).

TPB has been extensively utilized in diverse areas to explore and predict individuals' ITU technology or perform specific behaviors in particular contexts. These applications include educational technology (Al-Mamary et al., 2024; Nguyen et al., 2022), and AI technologies or tools (Al-Qaysi et al., 2025; Li, Jiang, et al., 2022). For example, in the domain of AI technologies, Li, Jiang, et al. (2022) used TPB to investigate medical students' attitudes and ITU AI technologies in clinical practice. Their unified model clarified 86.0% of the total variability in ITU.

To better adapt TPB to various contexts, researchers have increasingly integrated TPB with other theoretical models, such as TAM and Expectation Confirmation Theory (Jiao & Cao, 2024; Leou & Wang, 2023; Pan, Pan, et al., 2024; Wang et al., 2022; Zhao et al., 2024). For instance, Zhao et al. (2024) incorporated the Technology Readiness Index, TAM, and TPB to examine determinants shaping university students' acceptance of ChatGPT, demonstrating that SN and BA were instrumental in determining their ITU ChatGPT. In summary, TPB and its extensions provide effective tools for analyzing BA and SN, enabling the understanding and prediction of human behavior across disciplines. Its applicability continues to expand in various fields, offering valuable insights into behavioral intention.

2.1.3. Digital literacy

DL involves the skill set required to navigate digital spaces, including the comprehension, critical assessment, effective application, and generation of content (DIGCOMP, 2013). In educational contexts, DL includes the ability to showcase creativity, curiosity, exploration, collaboration, and critical thinking using digital technologies (Lim, 2023). As AI technologies continue to develop, more and more educational tools have begun to integrate AI components into their design. This development demands that teachers not only possess basic information technology skills but also exhibit advanced DL to make the most of the possibilities offered by these tools (Wu et al., 2024; Zhang & Zhang, 2024). DL plays a vital role in enhancing the effectiveness of teaching and learning and in preparing students to succeed in today's rapidly evolving world (Damanik & Widodo, 2024).

Previous studies, often based on TAM and TPB, have explored the impact of DL on users' intentions to adopt new technologies (Feng & Sumettikoon, 2024; Liang et al., 2024; Lu, Sun, et al., 2024b; Ullah et al., 2022). As an illustration, Ullah et al. (2022) investigated the direct and indirect effects of DL on individual banking users' ITU mobile payment services, revealing that DL significantly and positively influenced ITU, both directly and indirectly through PEOU. Lu, Sun, et al., 2024b, based on TPB, found that DL enhanced farmers' SN and BA, increasing their likelihood of engaging in pro-environmental behaviors. Feng and Sumettikoon (2024) integrated TPB and TAM to examine DL's connections with additional factors, demonstrating the model's applicability across different contexts. In addition, studies have shown that teachers' levels of DL significantly and positively influence their PU, PEOU, attitudes, and ITU AI-assisted teaching tools (Mailizar et al., 2022; Nja et al., 2023; Yao & Wang, 2024). Yao and Wang (2024) investigated the key factors that shape pre-service special education teachers' intentions to incorporate AI into their prospective classroom practices, revealing that DL significantly and positively impacts PU and PEOU, which subsequently

influence teachers' ITU AI-powered teaching tools. Therefore, it is essential to further explore in greater depth how DL influences teachers' ITU AI tools into their instructional methods.

2.2. Research hypotheses

2.2.1. DL, PU, PEOU, and ITU

Numerous studies have confirmed that DL exerts a marked and beneficial effect on PU, PEOU, and ITU (Hwang et al., 2023; Nazzal et al., 2021; Ullah et al., 2022; Wang & Li, 2024; Yao & Wang, 2024). Yao and Wang (2024) examined pre-service special education teachers and found that DL positively influenced PU and PEOU, with PU mediating the link between DL and ITU. Ullah et al. (2022) revealed that DL significantly enhanced PEOU and ITU in mobile payment adoption.

Extensive research based on TAM has also confirmed the interrelationships among PU, PEOU, and ITU (Baca et al., 2023; Lu, He, et al., 2024; Nja et al., 2023; Tawafak et al., 2023; Zhang & Hou, 2024). For example, Nja et al. (2023) found that PEOU significantly influenced PU, which in turn shaped science teachers' ITU AI-assisted tools. Zhang and Hou (2024) reported that PU and PEOU significantly predicted university teachers' ITU AI teaching systems.

In light of the studies outlined above, this research hypothesizes that in the process of teaching with AI teaching tools, teachers who possess higher DL are more inclined to master these tools easily, enhancing their PEOU. This, in turn, increases their PU of the tools, thereby strengthening their ITU AI teaching tools. Accordingly, the following hypotheses are put forward:

H1: Teachers' DL significantly and positively influences their ITU.

H2: Teachers' DL significantly and positively influences PU.

H3: Teachers' DL significantly and positively influences PEOU.

H4: Teachers' PEOU significantly and positively influences PU.

H5: Teachers' PU significantly and positively influences their ITU.

H6: Teachers' PU mediates the relationship between DL and ITU.

2.2.2. DL, SN, BA, and ITU

A significant number of studies indicate that DL significantly and positively influences SN and BA (Cetindamar et al., 2024; Chen et al., 2024; Lu, Sun, et al., 2024b; Xu et al., 2024). For example, Xu et al. (2024) investigated determinants of users' willingness to spot inaccurate information on short video platforms, revealing that DL exerted a notable positive influence on SN. Similarly, Cetindamar et al. (2024) surveyed the link between employees' DL and their ITU digital technologies, revealing that enhanced DL fosters the development of positive BA.

Additionally, based on TPB, extensive research has confirmed the relationships among SN, BA, and ITU (Al-Qaysi et al., 2025; Jiao & Cao, 2024; Wang & Kang, 2024; Zhao et al., 2024). For instance, Al-Qaysi et al. (2025), in their investigation of the aspects influencing students' decision to use generative AI, found that BA are a significant driving factor, and SN also have a substantial impact on their ITU. Zhao et al. (2024) examined the determinants of university students' ITU ChatGPT, highlighting the importance of BA and SN in enhancing students' ITU ChatGPT.

In light of the studies outlined above, this research hypothesizes that higher levels of teachers' DL strengthen their SN and BA toward using AI teaching tools, which consequently enhances their readiness to use these tools for teaching purposes. In light of the above, this study formulates the following hypotheses:

H7: DL significantly and positively influences SN.

H8: DL significantly and positively influences BA.

H9: SN significantly and positively influence ITU.

H10: BA significantly and positively influence ITU.

H11: SN mediate the relationship between DL and ITU.

H12: BA mediate the relationship between DL and ITU.

2.2.3. DL, PU, BA, and ITU

As discussed earlier, previous studies have demonstrated that teachers' DL significantly and positively influences their PU (Nazzal et al., 2021; Ullah et al., 2022; Yao & Wang, 2024). Building on this finding, the present study further examines the feasibility of TAM and TPB in explaining teachers' technology acceptance.

TAM and TPB originate from the Theory of Reasoned Action. Their integration greatly enhances the explanatory and predictive power in understanding users' ITU AI tools (Lu, He, et al., 2024). Several studies integrating TAM and TPB have verified the favorable impacts effects of PU and BA on ITU (Adawiyah et al., 2024; Lu, He, et al., 2024; Prikshat et al., 2024; Xu & Deng, 2024; Zhao et al., 2024). For instance, Lu, He, et al. (2024), using an integrated model of TAM and TPB to investigate teachers' ITU generative AI technologies, found that PU significantly and positively influenced BA. Kodrat et al. (2024) examined the repercussions of DL on employees' ITU new technologies in the workplace, demonstrating that DL indirectly influenced ITU through PU and BA.

Drawing insights from previous work, this study puts forward the assumption that teachers with higher levels of DL will perceive AI teaching tools as more useful, which will positively shape their BA toward using these tools in teaching activities and subsequently enhance their ITU them. Based on this, the following hypotheses are put forward:

H13: Teachers' PU significantly and positively influences their BA.

H14: Teachers' PU and BA serve as a sequential mediator between DL and ITU AI teaching tools.

2.2.4. The moderating effects of gender and teaching experience

Against the backdrop of the widespread integration of educational technology, gender differences have gradually gained attention as a crucial element affecting teachers' ITU such tools. According to Li, Wang, et al. (2022), male teachers tend to highlight how technology contributes to improving teaching effectiveness, making PU a more significant predictor of their ITU. Conversely, women educators are often more prone to SN and external expectations, displaying an enhanced awareness of social dynamics. (Teo & Noyes, 2014; Venkatesh & Morris, 2000). Moreover, females are often more cautious when forming BA, which may enhance the predictive power of BA on their ITU (Teo & Noyes, 2014). Regarding DL, males generally exhibit greater confidence in their technological abilities, whereas females may experience lower levels of technology self-efficacy, potentially dampening their intention to adopt technology (Lazar et al., 2020; Lucas et al., 2021). Given these considerations, this study employs MGA to further explore whether gender moderates the effects of PU, BA, SN, and DL on teachers' ITU AI teaching tools.

In addition to gender, teaching experience – an important indicator of professional development – may also play a moderating role in shaping teachers' ITU AI teaching tools. Previous research has indicated that educators who have accumulated more years in the profession are more likely to be more conservative toward emerging technologies, displaying a weaker response to PU. In contrast, early-career teachers, being in the initial stages of professional growth, are more focused on improving teaching outcomes and thus place greater emphasis on the practical value of technology (Sahin et al., 2021). At the same time, BA, as manifestations of intrinsic motivation, may influence usage intentions differently depending on teaching experience. Individuals early in their teaching careers often embrace emerging educational technologies, largely influenced by their optimistic outlook, while those with extensive classroom experience tend to maintain conventional instructional strategies and exhibit diminished responsiveness to attitude-based motivators (Hou & Shen, 2024). Regarding SN, groups with different levels of teaching experience may vary in their perceptions of organizational culture, peer influence, and managerial expectations, potentially moderating the relationship between SN and ITU (Hou & Shen, 2024; Tarhini et al., 2014). Furthermore, teachers with shorter teaching experience are typically raised in more technologically developed environments, often possessing higher DL and technological confidence, which enhances their receptiveness to

technology-assisted instruction. As a result, DL may exert different effects on ITU across experience levels (Lucas et al., 2021; Peng et al., 2023). Therefore, this study adopts MGA approach to examine whether significant differences exist in the influence of PU, BA, SN, and DL on ITU, across groups with varying levels of teaching experience.

Based on this, the following hypotheses are put forward:

H15a: There are significant gender-based differences in the effect of PU on ITU.

H15b: There are significant gender-based differences in the effect of BA on ITU.

H15c: There are significant gender-based differences in the effect of SN on ITU.

H15d: There are significant gender-based differences in the effect of DL on ITU.

H16a: There are significant differences based on teaching experience in the effect of PU on ITU.

H16b: There are significant differences based on teaching experience in the effect of BA on ITU.

H16c: There are significant differences based on teaching experience in the effect of SN on ITU.

H16d: There are significant differences based on teaching experience in the effect of DL on ITU.

The hypothesized model for this study is presented in Figure 1.

3. Methodology

3.1. Sample and data collection

This study employed an online survey platform (https://www.wjx.cn/) to collect data from August to October 2024, aiming to validate the hypothesized model. To achieve a diverse and representative sample, participants were randomly chosen from five universities in Henan Province. Specifically, we first obtained a basic information list of teachers through the relevant departments of the universities. According to this list, a total of 500 teachers were identified across the relevant departments of the five universities. This list included data such as names, gender, age, and teaching experience but excluded any sensitive or personal information to protect teachers' privacy. Next, a computer-generated random number sequence was applied to proportionally and randomly draw a specified number of teachers from the roster of each university. Upon completion of the randomization, a total of 350 teachers were chosen to take part in the study. Each selected participant was sent an email invitation that outlined the study's objectives, importance, and procedures for participation. The email highlighted that participation was entirely voluntary and guaranteed that all personal data would be kept strictly confidential. Before starting the formal survey, participants were asked to fill out and submit the informed consent form via an online system, indicating that they comprehended the study's details and consented to take part. Of these, questionnaires were distributed to 350 teachers, and after eliminating incomplete or invalid responses, yielding 311 valid questionnaires in total. By

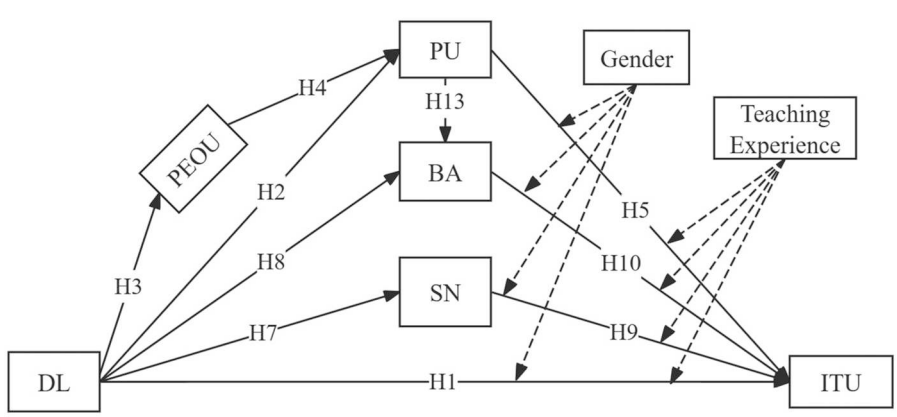

Figure 1. Research hypothesis model.

adopting this method, the study adhered to ethical research guidelines while enhancing the transparency and trustworthiness of the data collection method.

This study successfully collected 311 valid questionnaires. The maximum number of arrows pointing toward an endogenous latent construct in this study was three. According to Hair et al. (2014), to achieve a model explanatory power with an R^2 value of 0.10 at a 1% significance level, a minimum sample size of 91 is required. With 311 samples, this study significantly exceeded the minimum requirement, guaranteeing the validity and dependability of the outcomes. The demographic characteristics of the participants are as follows: 189 males (60.77%) and 122 females (39.23%), with the majority (161 participants, 51.77%) aged between 31 and 40 years. By teaching experience, 11 participants (3.54%) had less than three years, 177 (56.91%) had 3–15 years, and 123 (39.55%) had more than 15 years. By educational background, 211 participants (67.85%) held a master's degree, 76 (24.44%) a doctoral degree, and 24 (7.72%) were postdoctoral researchers. The analysis revealed significant associations with gender, age, teaching experience, and educational background ($p < 0.001$). The use of this representative sample forms a reliable groundwork for exploring the connections among teachers' DL, PU, PEOU, BA, SN, and ITU (Table 1).

3.2. Measurement tools

The survey included items that gathered participants' demographic information (e.g. gender, age, teaching experience) as well as self-reported data on the study constructs, including DL, PU, PEOU, SN, BA, and ITU. Validated scales were used to assess these constructs, and modifications were applied to the scales to adapt them to fit the study's context and aims. For instance, in the construct of PEOU, the original item, "I was able to balance the synergy between Intelligent Teaching Assistants and education," was modified to "I am able to coordinate the relationship between AI educational tools and teaching objectives," better reflecting participants' actual experiences within the specific research context. Similarly, the item "Learning through AI tools and systems for education is easy for me" was revised to "Using AI educational tools for teaching or learning is easy for me," making the question more specific and aligned with the research goals. These adjustments ensured the questionnaire content was more relevant to participants' experiences.

In addition to basic demographic information, the questionnaire includes key constructs such as DL, PU, PEOU, BA, SN, and ITU. Each construct is measured using a 5-point Likert scale, ranging from (1) "strongly disagree" to (5) "strongly agree." The scales for DL, PU, and PEOU were developed by Yao and Wang (2024). DL is designed to assess an individual's proficiency and ability to perform tasks across various digital platforms, consisting of 12 items (e.g. "I am knowledgeable about information technologies beneficial for professional learning") with a Cronbach's alpha of 0.928, indicating high reliability. PU measures the extent to which users believe AI teaching tools are beneficial for teaching and learning, comprising 4 items (e.g. "Using AI teaching assistants helps me learn and teach more effectively") with a Cronbach's alpha of 0.760, demonstrating good reliability. PEOU assesses the ease of learning and operating AI teaching tools, consisting of 4 items (e.g. "I find using AI teaching tools straightforward") with a Cronbach's alpha of

Table 1. Demographic characteristics of the sample.

Variable	Count	Percentage (%)	*P*-value
Gender			<0.001
Male	189	60.77%	
Female	122	39.23%	
Age			<0.001
21–30 years	28	9.00%	
31–40 years	161	51.77%	
41–50 years	93	29.90%	
51–60 years	26	8.36%	
61 years or above	3	0.96%	
Teaching Experience			<0.001
Less than 3 years	11	3.54%	
3–15 years	177	56.91%	
More than 15 years	123	39.55%	
Educational Background			<0.001
Master's Degree	211	67.85%	
Doctoral Degree	76	24.44%	
Postdoctoral	24	7.72%	

0.836, also indicating good reliability. The scales for BA and ITU were developed by Lu, He, et al. (2024). BA assess users' perceptions of using AI tools, comprising 3 items (e.g. "I am satisfied with using AI teaching assistants") with a Cronbach's alpha of 0.724, reflecting good reliability. ITU evaluates users' willingness to adopt the tools in the future, consisting of 3 items (e.g. "I am willing to use AI teaching assistants for teaching and research support") with a Cronbach's alpha of 0.745, indicating good reliability. The SN scale, developed by Hou and Shen (2024), measures the extent to which users perceive social influences regarding the use of AI teaching tools. It consists of 3 items (e.g. "Teachers at my school approve of my use of AI teaching assistants") with a Cronbach's alpha of 0.758, showing good reliability. Details of the measurement items are provided in Appendix A.

Two independent researchers translated the initial version of the questionnaire: one was a content expert fluent in psychology and the other a linguist with deeper expertise in English and Chinese. Subsequently, two other researchers retranslated the English version into Chinese. Both versions were contrasted, and any notable differences were addressed to form a consistent version. A pilot test of the final version was conducted with 20 university educators, and discrepancies or awkward phrasing compared to the initial English version were addressed, leading to the completion of the final Chinese version.

3.3. Data analysis

This study employed PLS-SEM to assess the validity of the framework and the relational model. PLS-SEM presents a range of significant benefits: first, it is a data analysis approach focused on prediction (Leguina, 2015); second, it proves effective with limited sample sizes and retains its robustness even when the data fails to conform to the normality assumption (Leguina, 2015); third, it effectively handles complex model structures (Premkumar & Bhattacherjee, 2008). Given these advantages, PLS-SEM was deemed suitable for this study.

The examination of the data focused on two key aspects. First, the assessment of the outer model, which evaluated the reliability and validity of the constructs. Reliability evaluation included item reliability, composite reliability, and Cronbach's alpha. Validity evaluation covered both convergent validity and discriminant validity. To verify discriminant validity, the Fornell-Larcker criterion and cross-loading tests were employed. Second, the evaluation of the inner model, which involved examining collinearity, testing the significance of model relationships, and assessing the explanatory power of the model (R^2).

4. Results

4.1. Common method bias

The data used in this study were all collected through self-reported measures, which may pose the risk of common method bias. To assess this issue, Harman's single-factor test was conducted. The results of the exploratory factor analysis revealed that three factors with eigenvalues greater than 1 were extracted, with the first factor accounting for 47.895% of the total variance, which is below the critical threshold of 50%. This indicates that there is no serious common method bias in this study (Podsakoff & Organ, 1986).

4.2. Measurement model

The examination of the measurement model entailed verifying the consistency and accuracy of the survey instrument and the collected data. Reliability indicators included the outer loadings of the items and the composite reliability of the scales. According to Byrne (2013), outer loadings greater than 0.70 indicate high reliability. Table 2 shows that Item 8 from the DL scale and Item 2 from the PU scale had outer loadings below 0.70, leading to the consideration of their removal. All other items had outer loadings exceeding 0.70, demonstrating high reliability.

To assess the scales' internal consistency, composite reliability and Cronbach's alpha was utilized as evaluation metrics. Table 2 indicates that the composite reliability for all scales ranged from 0.724 to 0.930, meeting the criteria of 0.7–0.9 (Hair, Hult, Ringle, Sarstedt, Danks, et al., 2021). Similarly, Table 2 shows that the Cronbach's alpha values ranged from 0.724 to 0.928, both exceeding the threshold of 0.7 as standard.

To evaluate convergent validity, the consistency among different indicators measuring the same construct was examined (Campbell & Fiske, 1959). Convergent validity was measured using the average variance

Table 2. Reliability and validity of the model.

Constructs	Items	Loadings	Cronbach's alpha	CR	AVE
DL	DL 1	0.705	0.928	0.930	0.582
	DL 2	0.736			
	DL 3	0.750			
	DL 4	0.742			
	DL 5	0.773			
	DL 6	0.716			
	DL 7	0.816			
	DL 9	0.790			
	DL 10	0.772			
	DL 11	0.781			
	DL 12	0.799			
PU	PU 1	0.829	0.760	0.760	0.676
	PU 3	0.822			
	PU 4	0.815			
PEOU	PEOU 1	0.800	0.836	0.838	0.671
	PEOU 2	0.806			
	PEOU 3	0.804			
	PEOU 4	0.863			
BA	BA 1	0.810	0.724	0.724	0.644
	BA 2	0.802			
	BA 3	0.795			
SN	SN 1	0.855	0.758	0.778	0.675
	SN 2	0.734			
	SN 3	0.869			
ITU	ITU 1	0.867	0.745	0.753	0.663
	ITU 2	0.778			
	ITU 3	0.794			

extracted (AVE). In this study, the AVE ranged from 0.582 to 0.676, exceeding the threshold of 0.50 (Fornell & Larcker, 1981).

Discriminant validity measures how clearly a construct is differentiated from other constructs (Zaiţ & Bertea, 2011), and it is typically assessed using the Fornell-Larcker criterion (Fornell & Larcker, 1981). Based on this criterion, the correlation coefficients between constructs should be lower than the square root of the AVE (Henseler et al., 2015). Table 3 indicates that the correlation values between the constructs fall beneath the square roots of their respective AVE values. Therefore, the results of this study meet the requirements for discriminant validity.

Introduced by Henseler et al. (2015), the Heterotrait-Monotrait Ratio (HTMT) acts as an additional measure for assessing discriminant validity.. Scholars generally recommend an HTMT ratio within the range of 0.85–0.90; a ratio exceeding 0.90 indicates excessive similarity between constructs, suggesting a lack of differentiation (Ab Hamid et al., 2017; Hair et al., 2014). As shown in Table 4, all HTMT values are below the critical threshold of 0.90 when evaluating the measurement model using the HTMT criterion. Thus, the findings satisfy the criteria for discriminant validity.

Table 3. Fornell-Larcker criterion.

	BA	ITU	DL	PEOU	PU	SN
BA	0.803					
ITU	0.632	0.814				
DL	0.694	0.749	0.763			
PEOU	0.632	0.663	0.751	0.819		
PU	0.643	0.621	0.724	0.674	0.822	
SN	0.645	0.635	0.742	0.678	0.639	0.822

Note: The bolded diagonal values represent the square root of the AVE.

Table 4. HTMT criterion.

	BA	ITU	DL	PEOU	PU	SN
BA						
ITU	0.859					
DL	0.845	0.897				
PEOU	0.812	0.836	0.850			
PU	0.866	0.822	0.858	0.846		
SN	0.866	0.834	0.876	0.840	0.828	

4.3. Structural model

4.3.1. Model fit evaluation

To assess the fit of the structural model, the standardized root mean square residual (SRMR) was examined. The SRMR value of the estimated model was 0.056, while that of the saturated model was 0.050. Both values fall below the recommended threshold of 0.08, indicating that the model demonstrates a good fit (Hair et al., 2022; Henseler et al., 2015).

4.3.2. Collinearity diagnostics

According to Hair, Hult, Ringle, and Sarstedt (2021), the Variance Inflation Factor (VIF) can be used to detect multicollinearity issues within a model. A VIF value below 3.3 is considered ideal, while values below 5 are deemed acceptable (Hair et al., 2019). As shown in Table 5, the VIF values for all variables span from 1.000 to 3.171, indicating that multicollinearity does not significantly affect the results of this study.

4.3.3. Path hypothesis testing

This section presents the results addressing Research Question 1, which investigates whether DL directly influences teachers' ITU AI teaching tools through the structural model path analysis.

During the examination of the structural model, significance testing is primarily used to determine whether the effects of exogenous variables on endogenous variables are statistically significant. The results from Table 6 and Figure 2 indicate the following: DL significantly and positively influences ITU ($\beta = 0.487$, $t = 6.429$, $p = 0.000$), PU ($\beta = 0.498$, $t = 7.507$, $p = 0.000$), PEOU ($\beta = 0.751$, $t = 21.723$, $p = 0.000$), SN ($\beta = 0.742$, $t = 21.987$, $p = 0.000$), and BA ($\beta = 0.481$, $t = 7.100$, $p = 0.000$). Thus, H1, H2, H3, H7, and H8 are supported. PU ($\beta = 0.095$, $t = 1.783$, $p = 0.037$), SN ($\beta = 0.108$, $t = 1.846$, $p = 0.032$), and BA ($\beta = 0.163$, $t = 3.198$, $p = 0.001$) significantly and positively influence ITU, supporting H5, H9, and H10. PEOU ($\beta = 0.300$, $t = 4.337$, $p = 0.000$) significantly and positively influences PU, supporting H4. PU ($\beta = 0.295$, $t = 4.433$, $p = 0.000$) significantly and positively influences BA, supporting H13.

4.3.4. Coefficient of determination (R^2) and predictive relevance (Q^2)

This section reports the findings for Research Question 2, which examines how well the proposed conceptual model accounts for the variation in teachers' ITU AI teaching tools.

R^2 measures how well independent variables account for the variance in dependent variables, thereby reflecting the overall effect of exogenous latent variables on endogenous latent variables. According to Chin (1998), R^2 values are often used to denote the strength of the relationship, with 0.67 indicating a

Table 5. Collinearity diagnostics for the structural model.

	BA	ITU	DL	PEOU	PU	SN
BA		2.206				
ITU						
DL	2.100	3.171		1.000	2.296	1.000
PEOU					2.296	
PU	2.100	2.338				
SN		2.460				

Table 6. Path hypothesis testing results.

Hypothesis	Path	β	t	p	Result
H1	DL→ ITU	0.487	6.429	0.000	Supported
H2	DL→PU	0.498	7.507	0.000	Supported
H3	DL→PEOU	0.751	21.723	0.000	Supported
H4	PEOU→PU	0.300	4.337	0.000	Supported
H5	PU→ ITU	0.095	1.783	0.037	Supported
H7	DL→SN	0.742	21.987	0.000	Supported
H8	DL→BA	0.481	7.100	0.000	Supported
H9	SN→ ITU	0.108	1.846	0.032	Supported
H10	BA→ ITU	0.163	3.198	0.001	Supported
H13	PU→BA	0.295	4.433	0.000	Supported

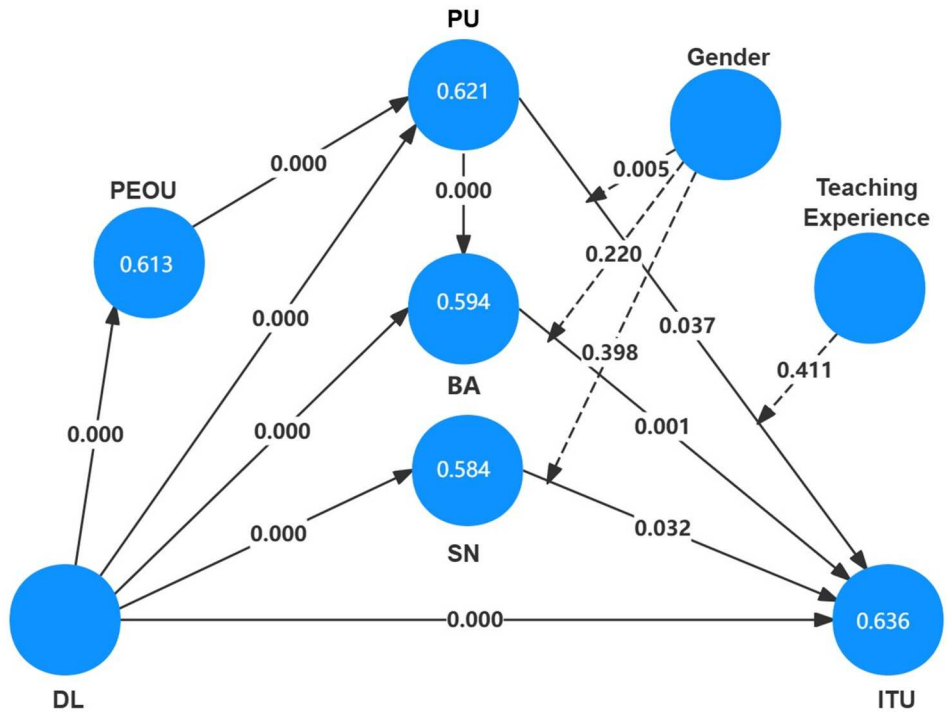

Figure 2. Path coefficients after bootstrapping.

strong interpretability, 0.33 representing a moderate one, and 0.19 signifying a weak one. In this study, the R^2 value for ITU is 0.595, indicating that the model explains 59.5% of the variance in the ITU, demonstrating a moderately strong explanatory power for the model.

However, as Hair et al. (2023) suggest, R^2 alone does not provide a comprehensive evaluation of a model's accuracy. As a result, the study implemented the Stone-Geisser Q^2 test, utilizing the blindfolding procedure to examine the predictive relevance of the structural model. All Q^2 values exceeded 0.35, indicating that the model demonstrates strong predictive relevance for the endogenous constructs (Chin, 1998). In this study, the Q^2 values were 0.476 for BA, 0.555 for ITU, 0.559 for PEOU, 0.517 for PU, and 0.545 for SN, demonstrating that the model exhibits predictive relevance and can effectively predict changes in the exogenous variables.

4.4. Mediating effect test

This section provides the results corresponding to Research Question 1 and Research Question 3, focusing on the mediating effects of key constructs in the relationship between DL and teachers' ITU AI teaching tools.

This study utilized the bootstrapping method in PLS-SEM to analyze mediating effects. Table 7 presents the mediating effects between different variables. DL showed a significant indirect effect on ITU through PU and BA ($\beta = 0.024$, $t = 2.477$, $p = 0.007$). While the direct effect was significant, the indirect effect accounted for less than 20% of the total effect, indicating a weak chained mediating role of PU and BA between DL and

Table 7. Analysis of mediating effects.

Relationship	Indirect Effect	*T*	*P*	Direct Effect	*T*	*P*	Type of Mediation
Mediation effect of BA							
PU→BA→ ITU	0.048	2.410	0.008	0.095	1.783	0.037	CPM
Indirect effects of DL							
DL→BA→ ITU	0.079	2.929	0.002	0.487	6.429	0.000	CPM
DL→PEOU→PU	0.225	4.191	0.000	0.498	7.507	0.000	CPM
DL→PU→BA→ ITU	0.024	2.477	0.007	0.487	6.429	0.000	CPM
DL→PU→BA	0.147	4.165	0.000	0.481	7.100	0.000	CPM
DL→SN→ ITU	0.080	1.813	0.035	0.487	6.429	0.000	CPM
DL→PU→ ITU	0.047	1.804	0.036	0.487	6.429	0.000	CPM

Note: NM – No Mediation; FM – Full Mediation; CPM – Complementary Partial Mediation.

ITU. Thus, H14 is supported. DL demonstrated a significant indirect effect on PU through PEOU ($\beta = 0.225$, $t = 4.191$, $p = 0.000$), while the direct effect was also significant, suggesting that PEOU plays a partially complete mediating role in this relationship. Additionally, DL showed significant indirect effects on ITU through BA ($\beta = 0.079$, $t = 2.929$, $p = 0.002$), SN ($\beta = 0.080$, $t = 1.813$, $p = 0.035$), and PU ($\beta = 0.047$, $t = 1.804$, $p = 0.036$). While the direct effect was also significant, the indirect effects accounted for less than 20% of the total effects, indicating relatively weak mediating roles. Therefore, H6, H11, and H12 are supported.

4.5. MGA results

4.5.1. Measurement invariance testing

This section presents the findings for Research Question 4, which investigates whether gender and teaching experience moderate the influence of DL on teachers' ITU AI teaching tools.

This research utilized MGA to examine and evaluate the variations in path coefficients between groups based on gender and levels of teaching experience. Such a method has been extensively applied in earlier academic studies (Zhou et al., 2014). Before exploring the moderating effects, this research applied the Measurement Invariance of Composite Models (MICOM) procedure to assess whether any concerns related to measurement invariance were present (Henseler et al., 2016).

According to the guidelines proposed by Hair et al. (2021a), conducting MGA requires establishing both configural invariance (i.e. identical model structure and estimation procedures) and compositional invariance (i.e. equal indicator weights). In Smart PLS 4.0, configural invariance is automatically ensured. Compositional invariance is assessed through a permutation algorithm (Hair et al., 2021a). In this study, the path models and data processing were identical across gender groups and teaching experience groups, which is a prerequisite for establishing configural invariance (Henseler et al., 2016). Additionally, the same algorithm settings were used in estimating the models, further supporting configural invariance (Henseler et al., 2016). If the correlation between construct scores exceeds the 5th percentile of the empirical distribution, compositional invariance is considered established (Henseler et al., 2016).

As presented in Table 8, the original correlations of the composite scores for each latent variable within the gender groups, except for DL, surpassed the 5% quantile of the empirical distribution, suggesting that compositional invariance was established for all other constructs. Therefore, full measurement invariance across gender groups was confirmed for all constructs except DL.

Regarding the teaching experience groups, due to a limited number of teachers with less than three years of experience (n = 11), the study focused on two more representative groups: those with 3–15 years and those with over 15 years of teaching experience, in order to better explore the impact of experience on teaching-related outcomes. As shown in Table 8, only the constructs PU demonstrated original correlations that exceeded the 5% quantile, satisfying the criterion for compositional invariance. However, the other latent variables – BA, ITU, DL, PEOU, and SN – did not pass the compositional invariance test. Therefore, measurement invariance across teaching experience groups is only partially supported, and full invariance cannot be established.

Table 8. MICOM step 2_compositional invariance.

Construct	Correlations among construct scores($H_0 = 1$)	5% Quantile of Empirical Distribution of c_u	*P* value	Compositional Invariance?
Gender				
BA	0.994	0.993	0.071	Yes
ITU	1.000	0.995	0.963	Yes
DL	0.999	0.999	0.044	No
PEOU	0.999	0.997	0.286	Yes
PU	0.998	0.996	0.281	Yes
SN	0.997	0.994	0.179	Yes
Teaching Experience				
BA	0.912	0.990	0.000	No
ITU	0.983	0.993	0.000	No
DL	0.993	0.998	0.000	No
PEOU	0.962	0.995	0.000	No
PU	0.995	0.994	0.065	Yes
SN	0.974	0.992	0.000	No

Table 9. Path coefficient comparisons.

	Original	Original	Original difference	2.5%	97.5%	*p*	Hypotheses
Gender	Male	Female					
BA →ITU	0.165	0.080	0.085	−0.191	0.179	0.220	Not Supported
PU →ITU	0.183	−0.094	0.278	−0.175	0.175	0.005	Supported
SN →ITU	0.110	0.088	0.022	−0.201	0.184	0.398	Not Supported
Teaching Experience	3–15 years	More than 15 years					
PU →ITU	0.088	0.072	0.017	−0.180	0.180	0.411	Not Supported

4.5.2. MGA

Following the confirmation of measurement invariance, MGA was conducted. To distinguish between gender and teaching experience groups, dummy variables were created to represent male vs. female and 3–15 years vs. more than 15 years of teaching experience (Venkatesh & Morris, 2000). The analysis was performed by comparing corresponding path coefficients across the different subgroups (Zhou et al., 2014).

The results of the path coefficient comparisons between gender and teaching experience groups are presented in Table 9. For the gender groups, the analysis revealed no significant differences in the effects of BA (β_male = 0.165, β_female = 0.080, $p = 0.220$), and SN (β_male = 0.110, β_female = 0.088, $p = 0.398$) on ITU. Therefore, hypotheses H15b, H15c, and H15d are not supported. However, a significant gender difference was found in the "PU → ITU" path (β_male = 0.183, β_female = −0.094, $p = 0.005$), indicating that PU has a significantly different impact on ITU between male and female teachers. Thus, H15a is supported.

As for the teaching experience groups, only PU met the criteria for compositional invariance. Therefore, the comparison focused on the "PU → ITU" path between the 3–15 years group and the over-15 years group. The results showed no significant difference between these groups (β_3–15 years = 0.088, β_>15 years = 0.072, $p = 0.411$). Accordingly, hypotheses H16a, H16b, H16c, and H16d are not supported.

5. Discussion

This study, grounded in TAM and TPB, explores the elements shaping teachers' ITU AI teaching tools, with a particular focus on the indirect effects of DL on ITU. Drawing upon an extensive review of existing literature, this research hypothesized that PU, PEOU, BA, and SN are strong indicators of teachers' ITU AI teaching tools, all of which are influenced by DL. Using PLS-SEM, the results revealed that DL and BA are the most critical predictors of ITU, congruent with the results of Al-Qaysi et al. (2025); Ullah et al. (2022), followed by PU and SN. DL significantly and positively affects PU, PEOU, SN, and BA, a finding supported by numerous studies(Lu, Sun, et al., 2024b; Nazzal et al., 2021; Ullah et al., 2022; Xu et al., 2024; Yao & Wang, 2024). Additionally, PU, SN, and BA play relatively weak mediating roles between DL and ITU, consistent with results from Lu, Sun, et al. (2024b); Yao and Wang (2024). There is also a weak chained mediation effect of PU and BA between DL and ITU, aligning with Kodrat et al. (2024).

This study demonstrates that DL has both direct positive effects on teachers' ITU AI teaching tools.

DL has a substantial positive effect on ITU, corroborating the findings of Ullah et al. (2022), who investigated the function of DL in shaping the ITU mobile payment systems among banking customers and found a similar positive influence. In this study, teachers with higher DL have a greater tendency to intend to use AI teaching tools. This may be because digitally literate teachers possess stronger technical and information-processing skills, enabling them to quickly grasp and apply AI teaching tools. They are more capable of comprehending the functions of the tools and application scenarios, integrating them into their daily teaching practices (Nurzhanova et al., 2024). Furthermore, improved DL enhances teachers' self-efficacy, helping them overcome fear and resistance to new technologies (Yao & Wang, 2024). Teachers with higher DL often exhibit greater perseverance and creativity, enabling them to handle challenges posed by new technologies more effectively (Damanik & Widodo, 2024). This not only increases their acceptance of technology but also strengthens their willingness to apply it in teaching practices (Rafique et al., 2024).

The proposed conceptual model effectively explains the variation in teachers' intention to use AI teaching tools, demonstrating strong predictive power. This study accounts for 59.5% of the variation observed in the ITU, with a Q^2 value of 0.555 for ITU, indicating strong predictive relevance of the model's predictors.

This study identifies PU, SN, and BA as key mediators in the relationship between DL and teachers' ITU AI teaching tools, although their mediating effects are relatively modest.

DL significantly and positively affects PU, which in turn strongly and positively affects ITU. Moreover, PU serves as a modest mediator between DL and ITU, consistent with the findings of Yao and Wang (2024). Their study explored the impact of teachers' DL on AI usage and demonstrated that PU serves as an intermediary in the connection between DL and the ITU. Teachers with stronger DL are better equipped to comprehend and utilize AI teaching tools, and they are able to precisely evaluate the instructional value these tools may offer. Consequently, their PU of AI teaching tools tends to increase (Yao & Wang, 2024). When teachers perceive that such technologies are beneficial to their work, their ITU AI teaching tools strengthens (Alkhawaja et al., 2022). Digitally literate teachers are also more capable of leveraging AI teaching tools to deal with real-world challenges in education, for example personalized learning and classroom management (Lozano & Blanco Fontao, 2023). Such advanced skills help teachers perceive the real value these tools contribute to their classroom practice, leading to heightened PU and ultimately fostering stronger ITU regarding AI teaching tools (Antonietti et al., 2022).

DL significantly and positively influences SN, which in turn significantly and positively affect ITU. Additionally, SN serves as a modest mediator between DL and ITU, consistent with the findings of Lu, Sun, et al. (2024b). Their research examined the connection between farmers' DL and their intentions to perform environmentally friendly actions, and found that SN serve as a mediator in the relationship between DL and pro-environmental behavioral intentions. Teachers who possess higher DL tend to be more inclined to engage in technological communication and collaboration with colleagues, enabling them to gain greater social support and resources (Akayoglu et al., 2020). This support includes not only technical assistance but also emotional encouragement and recognition. When teachers perceive positive attitudes toward new technologies from their surroundings – such as support from school management, mutual assistance among colleagues, and access to technical training – their SN are significantly strengthened (Molefi et al., 2024). Moreover, technological exchanges and collaborations among teachers allow achievements in using AI teaching tools by others to further enhance their own SN (Fan & Jiang, 2024). The more recognition individuals receive from their social circle or group, the higher their level of SN, which in turn strengthens their intention to adopt new technologies (Jiao & Cao, 2024). This trend is especially noticeable in educational settings, where teamwork and collective decision-making are integral to the work environment (Adelana et al., 2024). When teachers feel strong positive support from colleagues and management, they are more inclined to engage actively in the application and promotion of new technologies, ultimately enhancing overall teaching quality and efficiency (Adelana et al., 2024).

DL significantly and positively influences BA, which in turn significantly and positively affect ITU. Moreover, BA serves as a modest mediator between DL and ITU, aligning with the findings of Lu, Sun, et al. (2024b). Teachers with higher levels of DL often exhibit an open and curious mindset toward new technologies. Rather than feeling fear or resistance when encountering new tools, they adopt a proactive approach to exploration. This positive behavioral attitude further reinforces their affinity for AI teaching tools, encouraging them to actively integrate these tools within their instructional methods (Cetindamar et al., 2024). In an educational environment that encourages technological innovation and application, digitally literate teachers are more inclined to obtain support and recognition from colleagues and leaders. Such positive social feedback not only enhances their self-efficacy but also fosters the development of a positive behavioral attitude toward AI teaching tools (Nurhikmah et al., 2024). A positive behavioral attitude increases teachers' appreciation of AI teaching tools, encouraging them to integrate these technologies into their instructional planning and explore their application during real classroom activities. This proactive adoption ultimately promotes their ITU AI teaching tools.

PU and BA exhibit a weak chained mediating effect between DL and ITU, in accordance with the conclusions of Kodrat et al. (2024). Kodrat et al. (2024) investigated how information and DL impact employees' ITU new technologies in workplace settings, showing that DL influences ITU indirectly via PU and BA. This study shows that educators possessing stronger DL tend to regard AI teaching tools as valuable in boosting both teaching effectiveness and student performance, while also helping to make their instructional responsibilities easier and more efficient. These perceptions foster positive BA, encouraging teachers to adopt and experiment with these tools (Lu, He, et al., 2024). Furthermore, digitally literate

teachers are more inclined to engage in technological communication with colleagues. When they observe their peers successfully utilizing these tools, they feel inspired, which enhances their positive BA and willingness to adopt AI teaching tools (Gao et al., 2023). Teachers with high DL are also more likely to experience successful outcomes when applying AI teaching tools in practice, such as significant improvements in student performance and enhanced classroom interactions. These experiences further validate the tools' usefulness, reinforcing their positive attitudes (Adelana et al., 2024). Educators who possess positive BA toward AI teaching tools often demonstrate greater appreciation for these technologies and show a stronger tendency to incorporate them into their instructional practices (Gallego et al., 2024).

However, although PU, SN, and BA significantly intermediate the association between DL and ITU in this study – and PU and BA act as a chained mediator – their effects are relatively weak. These variables might be constrained by various external factors, such as the school's technological infrastructure, the level of managerial support, and the degree of teamwork among colleagues, which could affect teachers' ITU AI tools (Wang et al., 2024). Thus, in order to effectively encourage teachers to adopt AI teaching tools, it is essential not only to enhance DL, but also to refine the overall educational setting and offer comprehensive support and services. Schools should strengthen technological infrastructure, offer continuous professional development opportunities, establish supportive policy mechanisms, and encourage collaboration and knowledge-sharing among teachers. These measures would collectively facilitate the widespread application of educational technology and enhance teaching quality.

This study finds that gender show moderating effects on the relationship between PU and teachers' ITU AI teaching tools.

In the MGA, the path from PU to continued ITU exhibited a statistically significant difference across gender. Specifically, this path was significantly positive among male participants, while it was slightly negative among female participants. This finding indicates a substantial gender-based divergence in the importance placed on PU during the technology acceptance process. In particular, male users tend to base their judgments on instrumentality and practical benefits – that is, when a system is perceived as enhancing learning efficiency or task performance, their intention to continue using it increases (Li, Wang, et al., 2022). In contrast, for female users, ongoing ITU is shaped by a wider array of factors beyond PU, including emotional experiences, interface usability, interactivity, and social perceptions (Teo & Noyes, 2014; Venkatesh & Morris, 2000). Even if women recognize a system's practical value, shortcomings in user experience may still hinder their continued use or even trigger resistance or aversion (Lucas et al., 2021; Teo & Noyes, 2014). Therefore, to promote sustained use among female users, it is insufficient to merely enhance system functionality or efficiency; it is also crucial to address user experience, emotional engagement, and social support in order to boost overall acceptance and usage intention.

Moreover, the MGA revealed no significant differences in the effects of BA, SN, and DL on ITU between male and female teachers. Similarly, no significant differences were observed across teaching experience groups (3–15 years vs. over 15 years) in terms of the influence of PU, BA, SN, and DL on ITU. These findings may result from a confluence of factors. First, modern educational systems offer a relatively uniform professional environment and ongoing development opportunities to all teachers, regardless of gender or teaching experience. This ensures equitable access to training and resources, thereby minimizing disparities that might otherwise arise from individual background differences (Sánchez-Prieto et al., 2017; Walker et al., 2020). Second, the widespread culture of resource sharing and professional collaboration allows less experienced teachers to develop rapidly, while experienced teachers can also benefit from new technologies, fostering collective advancement across the teaching workforce (Alfadda & Mahdi, 2021). Additionally, individual characteristics – such as openness to technology and adaptability – may have a greater influence on ITU than gender or teaching experience. This suggests that personal interest and learning attitudes may overshadow intergroup differences (Kauttonen et al., 2025). In conclusion, the availability of supportive policies and the extensive spread of educational technology have created the conditions and incentives necessary for teachers to integrate new approaches and tools into their teaching (Walker et al., 2020). In summary, these findings reflect the combined impact of multiple factors, including the consistency of educational environments, equity in professional development, the role of individual traits, cultural dynamics, and policy support. Further research is required to examine the specific mechanisms underlying these effects in different contexts.

5.1. Impact

5.1.1. Theoretical implications

First, this study systematically explored the multi-layered influence mechanisms of DL – a core variable – on teachers' ITU AI teaching tools, incorporating four key mediators: PU, PEOU, SN, and BA. The results reveal that DL not only directly affects teachers' ITU but also exerts indirect effects through PU, SN, and BA. This finding transcends the traditional focus on the direct effects of DL by delving deeper into its multi-level indirect mechanisms. It provides a robust theoretical basis for educational technology adoption theories and enhances our comprehension of the crucial role of DL in technology acceptance.

Second, this study innovatively integrated the strengths of TAM and TPB, analyzing how DL influences teachers' ITU through different pathways from a multi-theoretical perspective. This cross-theoretical approach not only overcomes the limitations of relying on a single theoretical framework but also validates the chained mediating effect of PU and BA. Furthermore, it emphasizes the synergistic relationship between TAM and TPB in explaining complex behavioral intentions, offering valuable insights for future research.

Third, the inclusion of MGA based on gender and teaching experience advances the theoretical discourse on contextual moderators within technology acceptance models. Although most path relationships did not significantly differ across subgroups, the discovery of gender-based differences in the "PU → ITU" path highlights the importance of considering demographic variability in future theoretical frameworks. This supports the ongoing refinement of TAM and TPB models to incorporate user diversity, particularly in professional and pedagogical settings.

5.1.2. Practical implications

First, the findings highlight that teachers' DL is a key determinant in influencing their ITU AI teaching tools. Schools and educational management agencies should prioritize and increase investments in developing systematic and professional curricula tailored specifically for teachers. These AI education programs should cover comprehensive content, ranging from foundational concepts to advanced applications, accompanied by practical strategies and teaching case studies. Online educational platforms, including MOOCs, can be effectively utilized to provide teachers with flexible and diverse learning opportunities, ensuring they can access the latest knowledge and technological developments anytime and anywhere. Additionally, hands-on training should be emphasized by organizing regular workshops and practical sessions led by field experts. Such initiatives ensure teachers gain practical experience and application skills, significantly enhancing the efficiency of teaching processes and enriching students' learning experiences, thereby improving overall educational quality.

Second, the study emphasizes the significance of BA and SN in teachers' adoption of AI teaching tools. Building a supportive campus culture for integrating technology into teaching becomes essential. Schools can foster this culture by organizing technology-sharing events and establishing collaborative teacher communities where educators can share successful case studies and experiences in applying AI teaching tools. Such efforts can create positive role models and promote the dissemination of new technologies. Moreover, implementing policies that provide material or moral incentives – such as recognizing teachers who excel in technology integration – can further boost teachers' enthusiasm and acceptance of these tools.

Finally, the findings suggest that schools should pay attention to teachers' professional development and psychological support. By ensuring sustained professional learning opportunities, educators can better overcome hesitation and apprehension about adopting new technologies, which in turn boosts their self-assurance and increases their ITU regarding AI teaching tools. Establishing support systems, such as a "Teacher Technology Advancement Fund," can enable educators to participate in high-level AI education technology training and academic exchange programs both domestically and internationally, expanding their skills and perspectives. Additionally, schools should implement feedback mechanisms to collect teachers' concerns and suggestions regarding the use of AI teaching tools. This ongoing feedback can help optimize teaching tools and services. By implementing these comprehensive measures, schools can effectively enhance teachers' willingness and ability to adopt AI tools, driving the widespread adoption and application of educational technologies.

5.2. Limitations and future research recommendations

While this research sheds light on teachers' ITU AI teaching tools, there are several constraints that future research should address. Firstly, the research sample was limited to university teachers from specific regions in China, namely Henan Province, which reduces the applicability of the findings to various educational contexts, cultural backgrounds, and academic disciplines. Subsequent studies should broaden the participant base to involve teachers from multiple educational tiers and diverse global contexts, thus broadening the extent to which the findings can be applied. Secondly, a cross-sectional approach was utilized in the study, which, while effective in validating significant relationships between variables, fails to capture the dynamic changes and causal relationships between DL and ITU. Future research should adopt longitudinal designs to better capture these dynamic processes and provide stronger empirical evidence. Finally, this study relied solely on participants' self-assessments, which could introduce inherent personal biases. Future research should consider employing a combined approach that integrates both qualitative and quantitative methods. Utilizing approaches such as qualitative interviews, behavioral observations, or digital tracking systems could enhance the accuracy and completeness of the collected data.

6. Conclusion

Amid the swift progress of AI technology, tools such as intelligent teaching assistants are gradually transforming traditional teaching models and profoundly impacting teachers' instructional practices. This study integrated TAM and TPB to systematically explore the influence mechanisms of DL on teachers' ITU AI teaching tools. The findings reveal that DL not only directly and significantly impacts ITU but also indirectly influences it through the mediating roles of PU, BA, and SN. Furthermore, DL exerts an additional impact through the chained mediation of PU and BA. Through MGA, this study found that the path from PU to continued ITU exhibited a significant difference across gender. These findings offer a theoretical basis for advancing and implementing AI-powered educational technologies and deliver valuable practical suggestions for policymakers and educational administrators in areas such as teacher training and technical support.

Authors' contribution

Conceptualization: Yun Du; Methodology: Yun Du; Formal analysis and investigation: Yun Du; Writing – original draft preparation: Yun Du; Writing – review and editing: Yun Du; Supervision: Yun Du. All the authors have read and agreed to the published version of the manuscript.

Ethics approval

The researchers confirms that all research was performed in accordance with relevant guidelines/regulations applicable when human participants are involved (e.g. Declaration of Helsinki or similar). This study was approved by the Ethics Committee of Henan Women's Vocational College.

Informed consent

The participants received oral and written information and provided written informed consent before participating in the study.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Yun Du, born in Henan, China, in 1980, obtained her bachelor's degree from Zhengzhou University of Aeronautics in 2004. She pursued her Master's degree at Wuhan University from 2007 to 2010, successfully graduating in 2010.

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Data availability statement

The data that support the findings of this study are available on request from the corresponding author.

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Appendix

Reverence	Constructs	Constructs	Initial Question	Item
Yao and Wang (2024)	DL	DL 1	I know about the Information and Communication Technology resources that enhance the mastery in my subject.	I am knowledgeable about information technology that is beneficial for professional learning.
		DL 2	I know how to use digital information to design teaching methods in my subject area.	I am knowledgeable about information technology that is beneficial for subject teaching.
		DL 3	I know how to inspire my pupils to learn using Information and Communication Technology resources.	I am knowledgeable about information technology that helps attract students' attention.
		DL 4	I know of various Information and Communication Technology resources for that supports my professional development.	I am knowledgeable about information technology that promotes professional development.
		DL 5	I am using appropriate software to improve my skills in my subject.	I am able to use appropriate software to learn professional knowledge.
		DL 6	I am applying Information and Communication Technology resources in designing my teaching methods for my subject	I use information technology to design my subject teaching.
		DL 7	I can use Information and Communication Technology resources to inspire my pupils to learn in my lessons.	I am able to use information technology to engage my students.
		DL 8	I am using Information and Communication Technology resources to enhance my professional development.	I use information technology to promote my professional development.
		DL 9	It is important to use Information and Communication Technology to design instructional methods for me.	I believe it is important to use information technology resources to design my subject teaching methods.
		DL 10	It is important to me using Information and Communication Technology resources to inspire my pupils to learn.	I believe it is important to use information technology resources to motivate my students to learn.
		DL 11	It is important for me to communicate pupils' learning with Information and Communication Technology.	I believe information technology can help me understand students' learning status.

(*Continued*)

Continued.

Reverence	Constructs	Constructs	Initial Question	Item
		DL 12	It is important to me to use Information and Communication Technology resources to enhance my professional development.	I believe it is very important to use information technology resources to promote my professional development.
Yao and Wang (2024)	PU	PU 1	Using artificial intelligence tools and systems and systems for education helps me to learn more efficiently.	Using AI teaching assistants can help me learn and teach more effectively.
		PU 2	Studying with artificial intelligence tools and systems for education would improve my study efficiency.	Using AI teaching assistants can improve my learning and lesson preparation efficiency.
		PU 3	Artificial intelligence tools and systems for education are advantageous for my learning.	AI teaching assistants are beneficial for my learning and teaching.
		PU 4	Overall, I think that artificial intelligence tools and systems for education are useful in my daily study.	Overall, I believe AI teaching assistants are useful for my daily learning and teaching.
Yao and Wang (2024)	PEOU	PEOU 1	It is easy for me to learn to use Intelligent Teaching Assistants for education	I find it easy to learn how to use AI educational tools.
		PEOU 2	Leaning though Intelligent Teaching Assistants for education is easy for me.	Using AI educational tools for teaching or learning is easy for me.
		PEOU 3	It is easy for me to become proficient in using Intelligent Teaching Assistants for education.	I am able to master the use of AI educational tools proficiently.
		PEOU 4	Overall, I think that Intelligent Teaching Assistants for education are easy to use.	Overall, I find AI educational tools easy to operate.
Lu, He, et al. (2024)	BA	BA 1	I am satisfied with the use of Intelligent Teaching Assistants	I am satisfied with using AI teaching assistants.
		BA 2	I support the use and promotion of Intelligent Teaching Assistants in my school.	I support the use and promotion of AI teaching assistants in schools.
		BA 3	I support regional governments in advancing Intelligent Teaching Assistants practices on the ground	I support the promotion of AI teaching assistants by local governments in practice.
Hou and Shen (2024)	SN	SN 1	School teachers agree with my use of Intelligent Teaching Assistants	School teachers approve of my use of AI teaching assistants.
		SN 2	School leaders support my use of Intelligent Teaching Assistants	School leaders support my use of AI teaching assistants.
		SN 3	Students are receptive to my use of Intelligent Teaching Assistants	Students are accepting of my use of AI teaching assistants.
Lu, He, et al. (2024)	ITU	ITU 1	I am willing to use Intelligent Teaching Assistants for assisted teaching and research	I am willing to use AI teaching assistants to support teaching and research.
		ITU 2	I am willing to incorporate Intelligent Teaching Assistants for individualized instruction	I am willing to adopt AI teaching assistants to implement personalized teaching.
		ITU 3	I am willing to utilize Intelligent Teaching Assistants for automated reviews	I am willing to use AI teaching assistants for automated grading.